

Worksheet — Angled Launch Simulation

Simulation 12 · use with *Simulation - Angled launch.html* · A.1 Kinematics: projectile launched at an angle — components of u , maximum height, time of flight, range; goes with the launcher / water-jet range demo

Name: _____ Class: _____ Date: _____

How to use this worksheet

Open **Simulation - Angled launch.html** in a browser (it works offline). Set the **speed u** , the **angle θ** and the launch **height h** , then  Launch. With **Keep paths** on, earlier launches stay on the screen so you can compare angles.

Part 1 before the lab (≈ 15 min) · **Part 2** during the demo (≈ 15 min) · **Part 3** homework (≈ 30 min).

Use $g = 9.81 \text{ m s}^{-2}$. Resolve first: $u_x = u \cos \theta$, $u_y = u \sin \theta$. Then horizontal $a_x = 0$, vertical $a_y = -g$.

Part 1 — Pre-lab: predict, then test

Write your prediction **before** you run the simulation. Then test it and explain what you saw.

Question	My prediction (before)	What the simulation showed	Physics explanation
1. Same speed, from the ground: which angle gives the longest range?			
2. Compare 30° and 60° (same speed). How do the ranges compare?			
3. What is the velocity of the ball at the highest point?			
4. Which angle (15° , 45° or 75°) keeps the ball in the air longest?			

5. Calculate the time of flight for $u = 10 \text{ m s}^{-1}$ at 30° from the ground. Check with the simulation.

Part 2 — During the lab: the range demo

Demo. The launcher (or water jet) fires at the same speed at different angles; mark where it lands. **Simulation:** $u = 6.0 \text{ m s}^{-1}$ from the ground, same angles, Keep paths on, ● Record each launch.

$\theta / ^\circ$	R (demo) / m	R (simulation) / m	difference / %
15			
30			
45			
60			
75			

1. Range at 45° in the simulation ($u = 6.0 \text{ m s}^{-1}$). Show the calculation.

2. Which pairs of angles land at the same place? Explain using $R = u^2 \sin 2\theta / g$.

3. In the demo the longest range was 3.0 m, at about 45° . Estimate the launch speed of the demo.

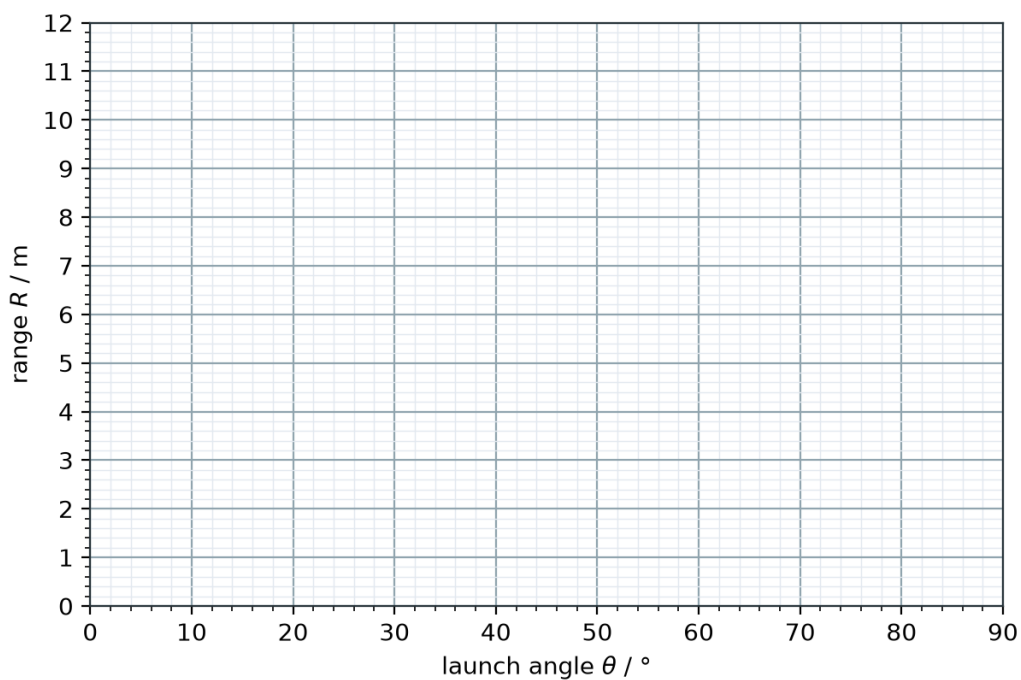
4. Why are the demo ranges a little shorter than the simulation, especially at steep angles? Give two reasons.

Part 3 — Homework: analyse

$u = 10 \text{ m s}^{-1}$ from the ground. Record one launch at each angle.

$\theta / ^\circ$	10	20	30	40	50	60	70	80
R / m								

1. Plot R (y-axis) against θ (x-axis) and draw a smooth curve.



2. Maximum range from your graph: $R = \underline{\hspace{2cm}} \text{ m}$ at $\theta = \underline{\hspace{2cm}} ^\circ$

3. A football is kicked from the ground at 15 m s^{-1} , 35° above the horizontal. Calculate the maximum height.

4. Calculate the time of flight.

5. Calculate the range.

6. What is the speed of the ball at the highest point?

7. A stone is thrown from the top of a 20 m cliff at 12 m s^{-1} , 30° above the horizontal. How far from the foot of the cliff does it land?

8. From the top of the cliff, is the best angle for the longest range 45° , more or less? Explain.
